On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues.

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①

$$\begin{pmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/2 & 1 \end{pmatrix}$$

$$\begin{matrix} 1 & \alpha & -\gamma & \alpha \\ 0 & 5-3\alpha^2 & -9\gamma-3\alpha & 1-3\alpha^2 \\ 0 & \beta-2\alpha & 3+2\gamma & \alpha^2-2\alpha \\ 0 & 14-\alpha & -6+\gamma & 2-\alpha \end{matrix}$$

$$\begin{matrix} 1 & \alpha & -\gamma & \alpha \\ \frac{3\alpha}{5} & 1 & -\frac{9\gamma}{5} & 1/5 \\ 0 & \beta-2\alpha & 3+2\gamma & \alpha^2-2\alpha \\ 0 & 14-\alpha & -6+\gamma & 2-\alpha \end{matrix}$$

$$\begin{matrix} 1 & 14 & -6 & 2 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \end{matrix}$$

$$\begin{matrix} 1 & 14 & -6 & 2 \\ 1 & \beta/2 & 3/2 & \alpha^2/2 \\ 0 & \alpha-14 & -\gamma+6 & \alpha-2 \\ 0 & 5-42\alpha & -9\gamma & -6\alpha \end{matrix}$$

$$\begin{matrix} 1 & 14 & -6 & 2 \\ 1 & \beta/2 & 3/2 & \alpha^2/2 \\ 0 & \alpha-14 & -\gamma+6 & \alpha-2 \\ 0 & 5-14,3\alpha & -9\gamma+6,3\alpha & -6\alpha \end{matrix}$$

$$\begin{matrix} 3\alpha-3\alpha \\ 0 \end{matrix}$$

(2)

$x = 5y$ intersection with another geometry volume?



(a) min dimension must be 3-d because volume exists at least 3-D.

(b) Geometry must contain 2 component. And must be independent from $x = 5y$. It can be a plane.

$x - 5y = 0$

$x + 4y - z = 6$

$$\begin{bmatrix} 1 & -5 & 0 \\ 1 & 4 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \end{bmatrix}$$

$C(A) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 5 \\ 4 \end{bmatrix}$ → all solutions inside $C(A)$ so intersection has 2-d inside $\mathbb{R}^2$.

rank = 2

(3)

H is symmetric matrix $H = I - 2uu^T$

$H = H^T$ because of symetriy

$H = I - 2uu^T = H^T$

$H \cdot H^T = I - 4uu^T + 4uu^T \cdot uu^T =$

$uu^T = 1$ $\quad uu^T = 1$

$H \cdot H^T = I$

$\downarrow$

$H^2 = I$

u is unit vector so $\|u\|^2 = 1$

(a) $H^2 = I$ can be shown directly.

(b) H is orthonormal → for orthonormality

$H \cdot H^T = I$

shown above.

(c)

$V_1 = u$

eigen vector.

$H = I - 2\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \Rightarrow$

$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ x \end{bmatrix}$

(d) diagonalizable !

④ $V_0 = 10$

$W_0 = 40$

$V_{k+1} = -V_k + W_k$

$W_{k+1} = V_k - W_k$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{pmatrix} V_k \\ W_k \end{pmatrix} =$$

$\det(A - \lambda I) = 0$

$$\begin{bmatrix} (\lambda+1) & -1 \\ -1 & (\lambda+1) \end{bmatrix}$$

$\lambda^2 + 2\lambda + 1 - 1$

$\lambda^2 + 2\lambda + 1 - 1 = 0$

$\lambda^2 + 2\lambda = 0$

$\lambda_1 = -2$

$\lambda_2 = 0$

$\lambda_1 = -2$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \alpha_1 = \begin{matrix} 0 \\ 0 \end{matrix}$$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} a_n \\ b_n \end{bmatrix} = c_1 \cdot \lambda_1 \cdot x_1 + \cancel{c_2 \cdot \lambda_2 \cdot x_2}$$

y_0

$$\begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} -2 \\ -2 \end{bmatrix} c_1$$

$1 = -2c_1$

$4 = -2c_1$